



Assessment Cover Sheet

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External Examiner's	Dr. Arun Arjunan		

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Learner ID	202101289	Date Submitted	25/05/2025
Learner Name	Abdulla Salaman		
Programme Code	ENT8020		
Programme	Bachelor of Engineering Technology (Mechanical)		
Lecturer's	Dr. Bilal Al-Elsayed, Dr. Ahmed Abdel-Rahman		
	Learner	Abdulla	

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Design and Development of locomotion design of a bio-inspired Soft Crawler Robot for Pipeline Inspection



EN8911 – ENGINEERING RESEARCH PROJECT

Submitted by

Abdulla Salamn

202101289

Supervised by

Dr. Bilal Al-Elsayed, Dr. Ahmed Abdel-Rahman

DEPARTMENT OF MECHANICAL ENGINEERING

SCHOOL OF ENGINEERING, BAHRAIN POLYTECHNIC

KINGDOM OF BAHRAIN

AUTHOR'S DECLARATION

I hereby declare that I am the sole author of this thesis. It is further declared that I have fulfilled all the requirements in line with the Quality Assurance guidelines of the Bahrain Polytechnic.

I understand that my report may be made electronically available to the public.

Signature:

Name: Abdulla Salman

Date: 25/05/2025

Abstract

This project presents the design and development of a bio-inspired soft crawler robot for pipeline inspection, utilizing pneumatic soft actuators to navigate confined spaces safely and efficiently. Traditional inspection methods often fall short in adaptability and can expose workers to hazardous environments. Our solution addresses these challenges by creating a lightweight, flexible robot capable of operating in pipes as narrow as 4 inches.

The robot's design leverages soft materials, inspired by earthworm locomotion, to achieve both radial gripping and linear movement. Key objectives include evaluating existing inspection methods, developing a modular robotic system, and integrating electronic controls for real-time monitoring. Rigorous testing and simulations confirm the robot's performance and reliability, demonstrating significant potential for enhancing pipeline inspection processes in Bahrain's oil and gas sector.

This work aligns with Bahrain's Economic Vision 2030, focusing on sustainable practices and innovative solutions. The successful implementation of this soft crawler robot represents a significant advancement in soft robotics, paving the way for future developments in automated inspection technologies.

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Chapter 1: Introduction

background

Pipelines are a critical part of modern infrastructure, transporting oil, gas, water, and industrial fluids. Over time, these pipelines face degradation due to corrosion, blockages, or material fatigue leading to costly maintenance, environmental hazards, and operational downtime. Traditional inspection methods, often relying on rigid robots or manual access, are limited in their ability to safely and efficiently navigate confined or curved pipeline environments.

Soft robotics presents a promising alternative, using flexible, compatible materials to achieve motion and adaptability being bio inspired by nature's earthworm. These systems offer lower risk of internal pipe damage and better navigation through variable geometries. This project introduces a low-cost, bio-inspired soft robot crawler designed to perform pipeline inspection using pneumatic soft actuators for both radial gripping and linear locomotion. The robot is specifically designed to address industry challenges in Bahrain and align with sustainable maintenance practices.

Problem statement

Most pipeline inspection robots in use today are rigid, complex, and limited to large-diameter pipelines. For instance, the commercially available FW100CT.2 robot is only compatible with pipes of 100 mm and above, making it unsuitable for narrower or more variable pipe systems commonly found in localized or aging infrastructure (RICO GmbH, n.d.). While the FW100CT.2 features advanced capabilities such as steerable wheels and integrated sensors, its size, complexity, and cost make it less accessible for low-budget operations or routine inspections. Furthermore, as highlighted by PipeDetect (n.d.), customers often face additional challenges with traditional robots, including high maintenance frequency, limited-service life, and complex operating procedures that require skilled users. These issues reduce efficiency, increase operational downtime, and make current solutions impractical for environments that require flexibility and low-cost implementation. Therefore, there is a clear need for a lightweight, adaptable, and damage-free inspection solution that can operate in smaller pipes (as small as 4 inches or 101.6 mm), requires minimal maintenance, and is easy to control. The proposed soft robot crawler addresses these needs using flexible, bio-inspired materials and pneumatic soft actuators, offering a non-disruptive, affordable, and user-friendly alternative to traditional rigid systems.

In addition to the limitations of current technologies, many pipeline inspections, especially in the oil and gas sector are still performed manually, exposing human workers to confined, hazardous environments. This method exposes workers to serious safety and health hazards, and in the unfortunate event of an accident, companies face severe consequences. These include not only the potential loss of life but also legal liabilities and substantial insurance payouts. According to global industry trends, workplace incidents in pipeline systems can cost oil companies millions in legal compensation and operational downtime, significantly impacting their financial and reputational standing (API, 2021). Furthermore, a detailed study by the Interstate Natural Gas Association of America (INGAA) found that pipeline testing and inspection costs can range from \$200 to over \$417 per linear foot, depending on the pipeline's location and risk classification contributing to an estimated \$1.3 billion annually (INGAA, 2016). These figures emphasize the urgent need for safer, automated, and cost-effective alternatives that minimize human exposure while maintaining inspection accuracy and regulatory compliance.

Project scope

This project focuses on the design, simulation, and analysis of a soft robot crawler for internal pipeline inspection. The robot's earthworm-like motion is bio-inspired. It utilizes pneumatic soft

actuators to achieve both radial expansion for gripping the pipe walls and linear extension for forward crawling.

The scope of the project includes:

- CAD modeling of the robot and its actuator components using SolidWorks
- Hyperelastic material simulation using ANSYS to analyze deformation, strain, and stress behavior
- Material selection for actuator casting based on flexibility, durability, and real-world performance
- Prototype development

Project objectives

The primary objective of this project is to design and analyze a soft robot crawler capable of performing safe, flexible, and efficient pipeline inspections using soft pneumatic actuators. The specific objectives are:

- To study the limitations of existing pipeline inspection methods and evaluate the potential of soft robotics for confined environments.
- To design a soft robot crawler model featuring radial and linear expansion mechanisms, inspired by earthworm-like locomotion.
- To develop detailed 3D CAD models of the robot using SolidWorks, focusing on actuator geometry and integration.
- To simulate the actuator's mechanical behavior using ANSYS, focusing on total deformation, elastic strain, and stress using hyperelastic analysis.
- To prototype selected parts (such as the linear actuator and radial balloon mechanism) using soft silicone casting.
- To assess the deformation performance and suitability of materials under pressure for crawling movement within 3-inch pipelines.

Bahrain's perspective

Bahrain's economy is heavily reliant on its oil and gas sector, with companies such as Bapco Energies operating massive underground pipeline networks across the Kingdom. These pipelines are vital for the transportation of crude oil and refined products, and their maintenance is essential to avoid leaks, corrosion, or costly downtime. Traditional pipeline inspection methods in Bahrain often involve manual labor or rigid robotic systems, which are limited in their adaptability and pose significant risks in narrow or aging pipelines. In the event of inspection failures or accidents, companies not only face operational delays but also high compensation and insurance liabilities.

The development of a low-cost soft robot crawler addresses a direct need in Bahrain's industrial context. Designed to navigate pipes as narrow as 3 inches, this soft robot offers a safe, flexible, and efficient alternative for internal inspection, especially in environments where traditional robots cannot operate. Its bio-inspired earthworm motion and use of soft pneumatic actuators ensure minimal damage to internal pipe walls and eliminate the need for human entry into confined, hazardous areas.

Moreover, this project aligns with Bahrain's Economic Vision 2030, which emphasizes sustainable development, innovation, and a shift toward a knowledge-based economy. It also pairs with Bapco's recent initiatives aimed at modernizing pipeline infrastructure, such as the new AB-4 pipeline project connecting Bahrain to Saudi Arabia. The proposed crawler could eventually support routine inspection, enhancing operational reliability while reducing long-term maintenance costs.

Chapter 2: Literature review

Yu et al. (2023): Design and Motion Simulation of a Soft Robot for Crawling in Pipes

Yu et al. (2023) presented a soft robot specifically designed for pipe crawling, incorporating both bending and elongation capabilities through modular actuator segmentation. The robot structure includes two deformable ends (front and rear) and a central expansion section, enabling inchworm-like locomotion. Each segment utilizes pneumatic actuation to either grip the pipe walls or extend the body, allowing the robot to crawl through confined spaces. The design was simulated using ABAQUS to test locomotion within pipelines of 20 mm to 42.4 mm diameter, including horizontal and curved sections.

The robot body was fabricated using silicone rubber and casted in mold assemblies, supporting soft behavior while maintaining resilience. The experimental validation confirmed smooth motion and strong adaptability in bent pipe environments. This study provides critical insights into this project's design of radial and linear actuators, particularly the integration of gripping and crawling. The performance validation via simulation also aligns with this project's approach, where hyperelastic material simulations are conducted in ANSYS to confirm deformation characteristics.

Key Contribution to Current Work:

- Validates the use of pneumatic chamber-based motion.
- Confirms soft robots' effectiveness in navigating tight, curved pipelines.
- Reinforces the feasibility of inchworm-like movement for confined inspection.

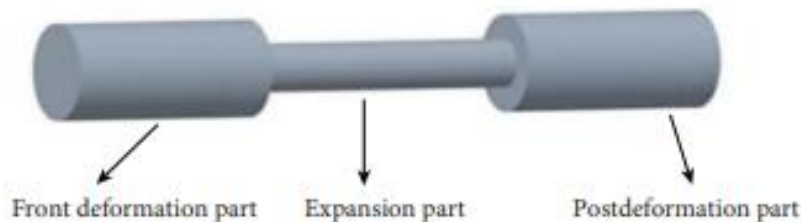


FIGURE 1: Schematic diagram of the pipe soft robot.

Figure 1 Diagram of the pipe soft robot.

Nguyen et al. (2024): Soft Robot Employing a Series of Pneumatic Actuators and Distributed Balloons

Nguyen et al. (2024) developed a soft pipeline inspection robot based on serial pneumatic actuators with embedded distributed balloon structures. These balloons serve as contact nodes, anchoring the robot against pipe walls during movement and enabling wave-like locomotion. The system operates via alternating pressure cycles, causing the actuator to steadily inflate and deflate along its length. The robot achieved a maximum forward velocity of 40.11 mm/s in horizontal settings and 9.22 mm/s in vertical climbing tests, showing high efficiency and controllability.

A key feature of the design is the modular configuration, allowing easy replication of actuator units for different lengths or strength requirements. The research also includes a theoretical model that estimates output force and motion velocity, verified through experimental data. This directly supports the development of the balloon-based radial actuator in this project, where internal inflation through radial grooves allows temporary anchoring against pipe walls.

Key Contribution to Current Work:

- Demonstrates effectiveness of balloon-based soft gripping systems.
- Offers a performance benchmark for radial actuator deformation and crawling speed.
- Supports modular actuator modeling and force estimation.

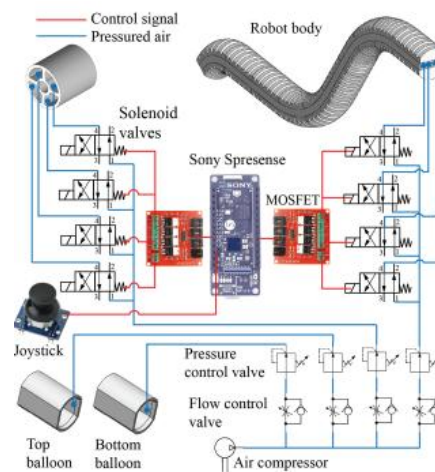


Figure 2 Work of soft robot.

Bar-Cohen (2014): Electroactive Polymers as Soft Robotic Actuators

Bar-Cohen (2014) provides a comprehensive review of electroactive polymers (EAPs) as a category of smart materials capable of producing large and flexible deformation under electrical stimulation. While the focus is not fully on pipeline inspection, the paper outlines the principles of soft actuator behavior, including energy efficiency, mechanical responsiveness, and material limitations. EAPs are shown to demonstrate muscle-like contractions and are proposed for applications requiring low weight, silent operation, and adaptability characteristics needed in soft pipeline robots.

Although the current project does not use EAPs directly (opting for pneumatic silicone-based systems), the study offers valuable context on material selection criteria, actuation behavior under stress, and the importance of safety and compliance in sensitive environments. It reinforces the need for low-voltage, high-deformation materials, which this project addresses through silicone actuators and pneumatic inflation.

Key Contribution to Current Work:

- Establishes soft actuation design principles.
- Highlights the importance of safe, flexible material deformation.
- Provides a broader context for selecting non-metallic actuator materials.

Chapter 3. Methodology

Overview

This chapter outlines the systematic approach to designing, fabricating, and testing a soft crawler robot designed for internal pipeline inspection. The development process involved the design and production of soft actuators using silicone casting, integration of a pneumatic actuation system, electronic control via Arduino, and a camera module for visual feedback. The goal was to create a functional, low-cost, and modular robotic system capable of adapting to confined pipeline environments.

Research

- **Literature Review:** Conduct a comprehensive review of existing research related to the project.

Product Development

- **Material Selection for Casting:** Select suitable materials for the casting process.
- **Circuit Component Selection:** Identify and choose the necessary components for the circuit.

Progress Evaluation and Feedback

- **Software Development:**
 - **Embedded System Programming (Arduino IDE):** Develop the firmware for the microcontroller.

Evaluation and Testing

- **Product Assembly and Integration:** Assemble the components and integrate them into the final product.
- **Component Testing and Calibration:** Test individual components and calibrate them for optimal performance.
- **Challenges and Limitations:** Identify and discuss any challenges encountered and limitations of the current design.

Research:

Literature Review

A comprehensive literature review was conducted to establish the context for the project, identify the need for efficient and adaptive pipeline inspection systems, and explore existing technologies and methodologies. The review covered the challenges associated with traditional rigid inspection robots in confined spaces, particularly within narrow or curved pipeline structures. Additionally, it focused on the emerging field of soft robotics and its potential to address these limitations through compliance, flexibility, and adaptability.

The literature also examined the role of pneumatic actuation in soft robotic movement and highlighted the advantages of using silicone materials such as Ecoflex for fabricating soft actuators. Key attention was given to low-cost fabrication techniques, including 3D printing and mold-based silicone casting, which have significantly reduced the barriers to prototyping and deploying soft robotic systems.

Furthermore, the integration of electronic control systems — such as Arduino microcontrollers and solenoid valves — was analyzed for its relevance to soft actuator operation. The importance of onboard camera systems in robotic inspection was also emphasized, particularly for providing visual feedback during internal navigation.

Finally, the findings from the literature review informed the design and fabrication of the crawler soft robot, ensuring that it addressed the specific functional requirements of low-cost, modular inspection in confined pipeline environments while maintaining simplicity, adaptability, and real-time monitoring capability.

Product Development

Material Selection for Liner and Connector:

The liner and connector components in the crawler soft robot serve as the critical interface between the soft actuator (balloon) and the pneumatic air supply. These parts must be flexible enough to accommodate actuator movement while also being durable, airtight, and structurally stable enough to withstand repeated inflation and deflation cycles.

To determine the most appropriate material for these components, several candidate materials were evaluated based on mechanical performance, moldability, and compatibility with the rest of the system.

Material	Flexibility	Tear Strength	Air Sealing Ability	Ease of Molding	Bonding with Silicone
Silicone Rubber	Moderate	High	Excellent	Good	Excellent
Ecoflex™ 00-30	High	Moderate	Good	Excellent	Excellent
Thermoplastic Polyurethane (TPU)	Moderate	High	Poor	Difficult	Poor
Latex Rubber	High	Low	Moderate	Good	Poor

Table 1 Material Selection for Liner and Connector.

Justification for Choosing Silicone Rubber:

The selection of **silicone rubber** for the liner and connector components was driven by its superior combination of mechanical strength, flexibility, and sealing performance. Unlike softer materials such as Ecoflex™ 00-30, silicone rubber offers greater tear resistance and structural integrity, making it ideal for components subjected to repeated pressurization and mechanical stress. Its excellent air sealing properties ensure that connections between the soft actuators and pneumatic tubing remain leak-free over multiple inflation and deflation cycles, which is critical for maintaining consistent actuator performance. Furthermore, silicone rubber is highly compatible with both Ecoflex™ 00-10 used in the balloon actuators and standard pneumatic tubing, allowing for reliable bonding at the interface points. From a fabrication perspective, silicone rubber is easy to mold using PLA molds, cures at room temperature without specialized equipment, and requires no additional surface treatments, aligning well with the project's low-cost and rapid prototyping goals. Taken together, these attributes made silicone rubber the most suitable material for the liner and connector components, balancing durability, airtightness, flexibility, and manufacturability.

Material Selection for Balloon Actuator:

Material	Flexibility	Elasticity	Tear Strength	Ease of Molding	Suitability for Pneumatic Expansion
Ecoflex™ 00-10	Very high	Excellent	Moderate	Excellent	Excellent – ideal for soft, large expansion
Dragon Skin	High	High	High	Excellent	Good – but slightly stiffer
Latex Rubber	Very high	High	Low	Moderate	Good – but poor durability over time
Silicone Rubber	Moderate	Moderate	High	Good	Limited – too stiff for balloon-like function

Table 2 Material Selection for Balloon Actuator

Justification for Choosing Ecoflex™ 00-10:

Ecoflex™ 00-10 was selected as the optimal material for the balloon actuator due to its exceptional softness, high elasticity, and proven suitability for pneumatic expansion in soft robotics applications. Its very low Shore 00-10 hardness allows it to undergo large, repeatable deformations without cracking or mechanical fatigue, making it ideal for creating soft actuators that rely on air pressure to generate movement. While alternatives like Dragon Skin offer excellent tear strength, they are stiffer and less compliant, which reduces the smoothness and range of actuator expansion. Latex

Circuit Component Selection

Component	Options Considered	Chosen Component	Reason for Selection
Microcontroller	Arduino Uno, ESP32, Raspberry Pi Pico	Arduino Uno	Easy programming, sufficient I/O pins, reliable for relay control, cost-effective, widely used in soft robotics
Relay Module	1-channel relay, 3-channel relay, MOSFET driver	3-Channel Relay Module	Provides independent switching of 3 valves, handles 24V load, integrates easily with Arduino, offers electrical isolation
Pneumatic Valve	12V solenoid, 24V solenoid, mechanical valve	24V Solenoid Valve	Higher switching performance, reliable air control, compatible with industrial-standard pneumatic systems
Power Supply	USB, 12V supply, DC 24V supply	DC 24V Power Supply	Delivers sufficient voltage and current to drive the 24V solenoid valves and air compressor reliably and continuously
Air Pump / Compressor	Mini diaphragm pump, piston compressor	24V DC Air Compressor	Provides required air pressure (100 kpa) for actuator inflation, compact, compatible with selected pneumatic system components

Table 3 Circuit Component Selection.

Justification for Selected Components

- Arduino Uno was selected as the central controller due to its simplicity, robust community support, and sufficient digital I/O capabilities for managing the relay signals. While more advanced boards like the ESP32 offer wireless capabilities, the Arduino Uno was sufficient for this application, which required only local relay switching.
- 3-Channel Relay Module was chosen to switch the 24V solenoid valves safely and independently. The relay module provides essential electrical isolation between the Arduino (low-voltage logic) and the high-voltage valve circuit, ensuring safe and reliable control.
- 24V Solenoid Valves were selected over lower-voltage alternatives because they provide faster response times, higher durability, and compatibility with standard pneumatic control systems. Mechanical or manually operated valves were rejected because they are unsuitable for automated robotic systems.
- DC 24V Power Supply was chosen to power both the solenoid valves and the 24V air compressor. This consolidated power approach ensures stable and sufficient current delivery, avoiding the limitations of USB-powered systems or lower-voltage supplies.
- 24V DC Air Compressor was selected for its ability to generate the necessary pneumatic pressure to inflate the soft actuators consistently. Its compact form factor and electrical compatibility with the system made it the best fit for the crawler robot's pneumatic subsystem.

Progress Evaluation and Feedback

- **Software Development:**

Embedded System Programming (Arduino IDE)

The firmware development for the crawler soft robot was carried out using the Arduino Integrated Development Environment (IDE), an open-source platform widely used for embedded system programming. The Arduino Uno microcontroller served as the central control unit, managing the timed actuation of the pneumatic system via relay-controlled solenoid valves.

Programming Design

The embedded system is designed with:

Microcontroller: Arduino Uno (R3)

Control Inputs: Serial commands via USB (open/close)

Outputs: Three digital pins (2, 3, 4) controlling three relays

At startup, the program initializes the relay pins as outputs and sets them to the deactivated (LOW) state. The Arduino listens for incoming serial commands: when the user types “open”, all three valves are activated (relays set HIGH), and when the user types “close”, all three valves are deactivated (relays set LOW).

Code Overview

```
const int relay1Pin = 2;
```

```
const int relay2Pin = 3;
```

```
const int relay3Pin = 4;
```

```
String command = "";
```

```
void setup() {
```

```
    pinMode(relay1Pin, OUTPUT);
```

```
    pinMode(relay2Pin, OUTPUT);
```

```
    pinMode(relay3Pin, OUTPUT);
```

```
    deactivateRelays();
```

```
    Serial.begin(9600);
```

```
    Serial.println("Type 'open' or 'close' to control relays.");
```

```
}
```

```
void loop() {
```

```
    if (Serial.available() > 0) {
```

```
        command = Serial.readStringUntil('\n');
```

```
        command.trim();
```

```
        if (command == "open") {
```

```

    activateRelays();

    Serial.println("Valves opened.");
} else if (command == "close") {
    deactivateRelays();

    Serial.println("Valves closed.");
} else {
    Serial.println("Invalid command. Use 'open' or 'close.'");
}
}
}

```

```

void activateRelays() {
    digitalWrite(relay1Pin, HIGH);
    digitalWrite(relay2Pin, HIGH);
    digitalWrite(relay3Pin, HIGH);
}

```

```

void deactivateRelays() {
    digitalWrite(relay1Pin, LOW);
    digitalWrite(relay2Pin, LOW);
    digitalWrite(relay3Pin, LOW);
}

```

This firmware setup provides a simple and effective command-based control system for managing pneumatic actuation in the crawler soft robot. Its modular design allows future upgrades, such as incorporating timed cycles, pressure sensor feedback, or even wireless command interfaces, to improve automation and autonomy.

Evaluation and Testing

Product Assembly and Integration:

The final stage of the fabrication process involved the systematic assembly of all mechanical, pneumatic, and electronic components into a fully integrated crawler soft robot system. This process was critical to ensure that each subsystem functioned reliably within the complete design, enabling coordinated motion, stable control, and real-time pipeline inspection.

Mechanical Assembly:

The final stage of the fabrication process involved the systematic assembly of all mechanical, pneumatic, and electronic components into a fully integrated crawler soft robot system. This process was critical to ensure that each subsystem functioned reliably within the complete design, enabling coordinated motion, stable control, and real-time pipeline inspection.

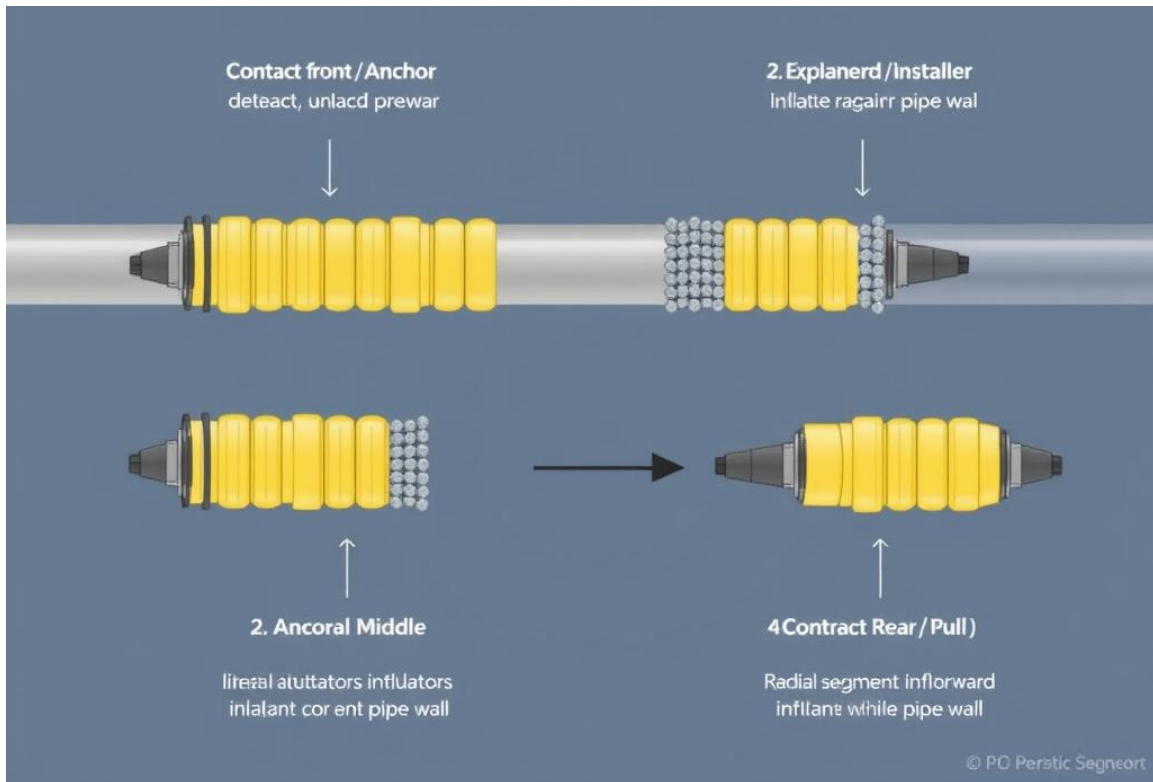
Liner Assembly:

Before assembling the components, the initial design of the liner underwent testing via casting processes, as illustrated.



Figure 3 First Liner Design.

The main issue with the casting process of the piece is the difficulty in extracting the central section, compounded by the presence of bubbles within the material. These bubbles compromise the piece's structural integrity and performance. The combination of challenging mold design and bubble formation leads to inefficiencies in production and potential failures in functionality.



We have conducted extensive research and engaged in creative design efforts to address the problem effectively. Our goal is to develop innovative solutions that enhance the casting process and improve the overall quality of the piece. We improved the design by incorporating a vacuum jumper, implementing significant changes to enhance the casting process. These modifications aim to reduce bubble formation and facilitate easier extraction, ultimately improving the overall performance of the piece.



Figure 4 Improve Design.

After developing the new shape, several improvements became evident. First, the design was altered to a square form, allowing for better stability during the casting process.

Additionally, the middle piece was separated from the main structure, which was 3D printed. Once silicone is poured into the mold, the middle pieces are positioned to create the desired shape, enhancing both the effectiveness of the cast and ease of extraction.



Figure 5 After pouring.

The problem depicted in the image relates to the casting mold used for producing the pieces. Several issues are evident, including the presence suggesting that air may have been trapped during the pouring process. This can lead to weak spots in the final cast product. Additionally, the uneven surface indicates that the material did not fill the mold uniformly, potentially affecting the overall quality and functionality of the resulting pieces. The complexity of the mold design may contribute to these challenges, making it difficult for the material to flow evenly and fill all sections properly.

Addressing these issues will be crucial for improving casting outcomes and ensuring the integrity of the final 'products'.



Figure 6 Finally Liner Design.

The solution involved separating the two pieces of the mold, which facilitates a smoother extraction of the cast components after pouring. This modification allows for the possibility of multiple pours without the risk of damaging the middle piece during extraction. Additionally, the design enhances the overall efficiency of the casting process, as the individual sections can be filled more effectively, reducing the likelihood of voids or bubbles. The image also highlights the improved mold structure, which features clearly defined shapes that can better accommodate the material flow, further ensuring the quality and integrity of the final cast products. This strategic redesign aims to address previous challenges while optimizing the production process.



Figure 7 Liner part.

The solution of separating the two pieces of the mold has proven highly effective, as evidenced by the significant reduction in bubble formation during the pouring process. The resulting cast components, as shown in the image, exhibit high quality with a smooth surface and no visible voids. This improvement indicates that the new design facilitates better material flow and ensures a more uniform fill, resulting in a more reliable final product.

Additionally, the clarity and consistency of the purple silicone pieces reflect the enhanced precision of the mold design. This not only improves the aesthetic appeal but also boosts the functional performance of the components. Overall, these advancements in the casting process highlight the importance of thoughtful design modifications in achieving high-quality outcomes.

To clarify, we performed the casting process twice using two similar pieces, and now we need to combine them into a final piece.



Figure 8 Assembly the part.

The process of aligning the two pieces on top of each other and adding material along the edges to bond them was quite complex. The service area presented some conflicting elements, making it challenging to achieve a perfect seal. Additionally, the expansion and contraction that occurred during the adhesion process added another layer of difficulty. These factors required careful calibration to ensure that the pieces would fit together seamlessly, minimizing gaps and inconsistencies. Despite these challenges, the effort was essential for creating a robust and effective mold that ultimately improved the quality of the final cast products.



Figure 9 Result of liner part.

While some pieces were successfully bonded, gaps remained due to the expansion and contraction during the process, as well as inconsistencies in the surfaces of the two pieces. This highlights the challenges faced, despite making progress towards forming the radiographic section. However, the experience gained from this attempt has led to a promising solution: utilizing a male and female mold design. This approach would allow for better alignment and bonding of the two pieces, minimizing gaps and ensuring a more seamless connection. By implementing this straightforward solution, we can enhance the effectiveness of the casting process and improve the overall quality of the final product.

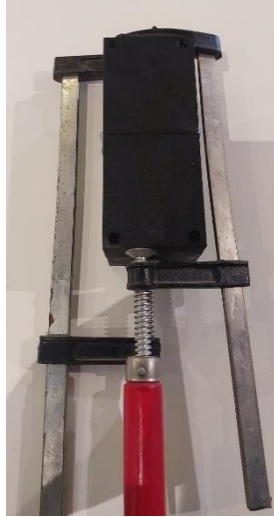


Figure 10 Assemble two part of liner.

By applying the male and female mold solution, we achieved a final result where the two pieces adhered seamlessly without any issues. This outcome represents the culmination of numerous changes and challenges faced throughout the process. We successfully created the final form of the radioactive piece, ensuring that the two components fit together perfectly without gaps. This significant improvement has enabled us to progress to the next stage of production, paving the way for enhanced efficiency and quality in our casting operations. The successful bonding of the pieces marks a pivotal moment in our development journey.

Connector Assembly:

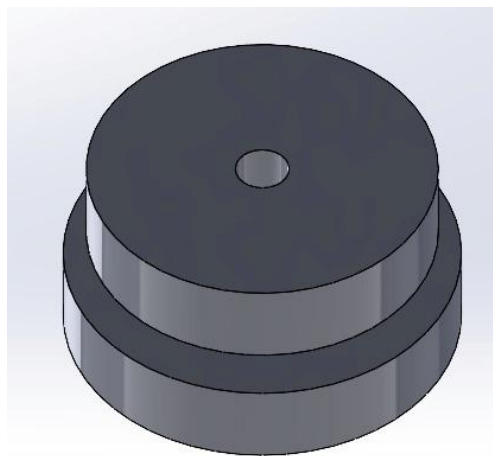


Figure 11 Connector Assembly.

The goal of the connector is to facilitate a reliable and seamless join between two components, ensuring structural integrity and functionality. By effectively bonding the pieces, the connector aims to eliminate gaps and inconsistencies that could compromise the overall performance of the final product. Ultimately, the connector serves as a critical element in achieving a high-quality, durable assembly.

Balloon Assembly:

We will review the stages as they occurred, and for additional details about the design mechanism and measurements, please refer to Fadel Ali's report.



Figure 12 First Design of Balloon.

As shown in the figure, the initial balloon design encountered several issues similar to those found in the liner, including the presence of bubbles and challenges during piece extraction. These problems hindered the overall quality and functionality of the product, necessitating a thorough assessment and redesign. Addressing these issues was crucial for ensuring a more effective casting process and improving the final outcomes.

After discussions with Fadhel Ali and the doctor, it was decided to make changes to the size of the balloon and to simplify the design. The new approach involves creating a structure

that can be closed from both the top and bottom, allowing for easier handling during the solidification of the radiation material. This modification aims to enhance the efficiency of the casting process and minimize issues related to bubbles and extraction, ultimately leading to a more effective final product.



Figure 13 Finally Design.

The primary challenge we faced was the density of the material, as the vacuum jumper proved ineffective in reducing the percentage of bubbles. After several attempts, we realized that the best approach was to allow the mixed material to sit for a few minutes, enabling the bubbles to rise to the surface. Once this was done, we could then use the vacuum jumper to further minimize the remaining bubbles. This method provided a more effective solution to address the issue, ultimately improving the quality of the final product.

Finall Assmbly:

The mechanical assembly began with the preparation of the soft balloon actuators, which were fabricated using silicone rubber. Once cured and demolded, the actuators were mounted onto the linear rail chassis to create the core crawling structure. The PLA-printed frame components provided alignment and support, ensuring the actuators were correctly positioned for sequential inflation-driven movement.

Components Assembled:

- Two Ecoflex balloon actuators
- Silicone rubber connectors and liners
- Camera module (mounted at the robot's front end)

Special care was taken to secure the actuators firmly to the rail system while allowing enough flexibility for expansion and contraction during actuation cycles.

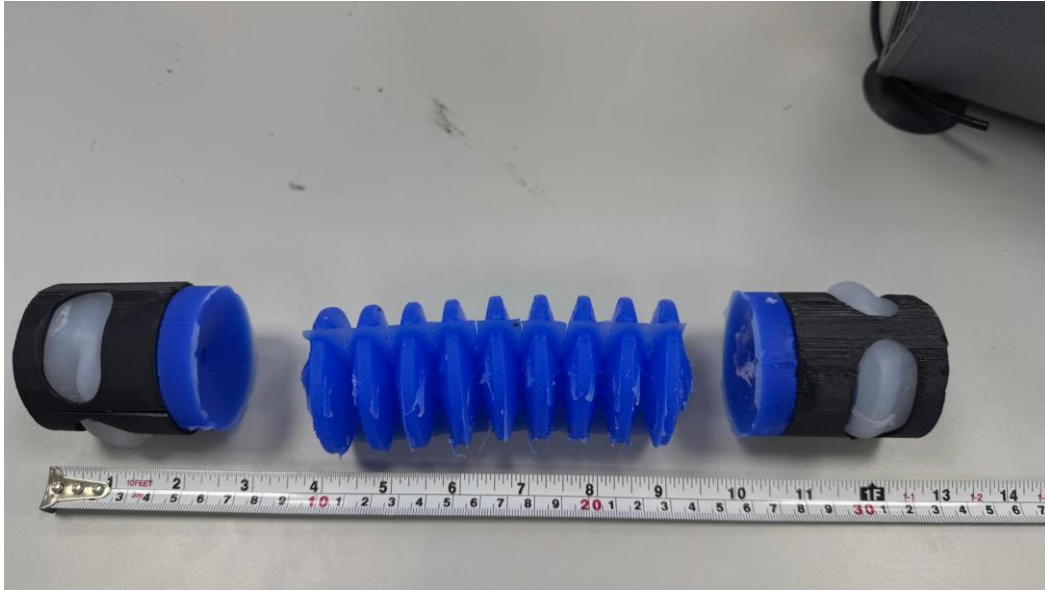


Figure 14 Soft Crawler Robot Part.

The goal is to ensure that each piece fits perfectly within the silicone material used to cast the liner. This will provide the necessary strength and durability to withstand the contraction and expansion cycles, particularly under air pressure conditions of



Figure 15 Soft Crawler Robot.

up to 100 kPa. By achieving this, we aim to enhance the overall performance and reliability of the components, ensuring they can effectively handle operational stresses during use.

The parts were successfully combined using silicone, as illustrated in the figure, without any issues during the assembly process. With the successful bonding of the components, the next step involves connecting the electrical circuit for the expansion process test. This

integration will allow us to evaluate the performance and functionality of the assembled pieces under operational conditions, ensuring they meet the required standards.

First, we connected three relays to three valves, ensuring that each valve is linked to its corresponding relay. The objective of this setup is to develop a program that simulates the sound process, allowing us to confirm that everything is functioning correctly before making the final connection. This pre-connection testing will help ensure that the system operates as intended and minimizes potential issues during the actual implementation.

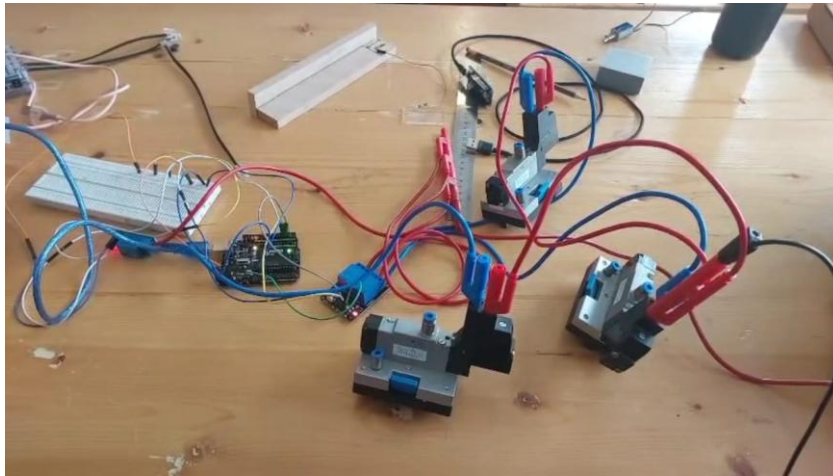


Figure 16 Test the Circuit.

The test yielded very positive results, with perfect connectivity and sound output from the Arduino. This confirmation allows us to proceed to the next stage, which involves connecting the connectors to the parts while utilizing the Arduino software for simulation. This step will ensure that the entire system functions as intended and supports the overall project goals effectively.



Figure 17 Final Circuit.

The connection process is now complete, and we are in the final stages. We will connect the system to the distributor, which is the compressor, and link it to the piece. Once everything is connected, we will run the code via the computer to observe the final results. This step is crucial for verifying the functionality and performance of the entire assembly.

Challenges and Limitations

In the development of the soft crawler robot for internal pipeline inspection, several challenges and limitations were encountered:

1. **Material Properties:** The selection of materials was critical. While silicone rubber was chosen for its durability and sealing capabilities, issues with density and air bubble formation during casting persisted. These bubbles compromised the structural integrity of the components, necessitating multiple iterations to achieve acceptable quality.
2. **Casting Process Complexity:** The casting process itself proved to be challenging due to difficulties in mold design and extraction. Bubbles trapped during the pouring process led to weak spots, affecting the overall performance of the cast pieces. Addressing these issues required significant redesign efforts and experimentation.
3. **Integration of Components:** Assembling the various components—soft actuators, pneumatic systems, and electronic controls—was complex. Ensuring compatibility and reliable connections among these systems required careful calibration and testing, which sometimes led to delays.
4. **Control System Limitations:** The embedded system programming, while functional, has limitations in terms of scalability and automation. The current setup relies on manual inputs for operation, which may not be ideal for all applications. Future iterations could benefit from more advanced control algorithms or wireless communication capabilities.
5. **Environmental Constraints:** The design must adapt to various pipeline conditions, which can vary significantly in size, shape, and material. This variability presents challenges in ensuring that the soft crawler can navigate effectively in all scenarios, potentially limiting its application.
6. **Testing and Evaluation Constraints:** The testing phase was limited by accessibility to real-world pipeline environments. Simulations provided valuable insights, but without extensive field testing, it is difficult to fully assess the robot's performance under operational conditions.
7. **Cost Considerations:** While the goal was to create a low-cost solution, the cumulative costs associated with material selection, prototyping, and testing can add up. Balancing performance with affordability remains a key challenge.

Chapter 4: Results and discussion

Chapter Overview:

This chapter presents the experimental, numerical, and analytical results derived from the development and testing of the soft crawler robot designed for internal pipeline inspection. It begins by revisiting the project's core objectives, focusing on the effectiveness of the soft actuators, the reliability of the pneumatic system, and the performance of the integrated electronic control.

Furthermore, the results underscore the practical implications of the soft crawler robot for various applications, particularly in improving pipeline inspection processes. The chapter also identifies the skill development needs required to further advance projects in this field, emphasizing the importance of interdisciplinary knowledge in robotics, materials science, and electronics.

Finally, the chapter addresses potential project risks, including material limitations and operational challenges, while discussing the sustainability and environmental impact of the developed solution. This comprehensive analysis serves to inform future research directions and enhancements in soft robotics technology.

Objective of the project:

This section revisits the core objectives established at the outset of the project, focusing on the development of a soft crawler robot for internal pipeline inspection. The primary goals include:

1. **Design of a Modular Robotic System:** To create a flexible and modular robot capable of adapting to diverse pipeline environments, ensuring efficient inspection in confined spaces.
2. **Integration of Soft Actuators:** To utilize soft actuators made from silicone materials, enhancing the robot's ability to navigate through narrow and curved pipelines while maintaining structural integrity.
3. **Pneumatic Actuation System:** To develop a reliable pneumatic system that enables precise control of the robot's movement through effective inflation and deflation of the actuators.
4. **Electronic Control via Arduino:** To implement an electronic control system using Arduino, facilitating real-time monitoring and operation of the robot, including the management of sensors and camera feedback.

5. **Performance Evaluation:** To test and evaluate the robot's functionality, including dispensing accuracy, sensor reliability, and overall efficiency in pipeline inspection tasks.
6. **Cost-Effectiveness:** To design a low-cost solution that remains accessible for various applications, while ensuring high-quality performance and durability.
7. **Sustainability Considerations:** To address environmental impacts by utilizing materials and processes that minimize waste and promote sustainable practices in the construction and operation of the robot.

Analytical Results

Test Methodology:

The results for the radiographic piece were outstanding, achieving perfect expansion that closely matched the simulation outcomes. The simulation indicated a 70% expansion rate, as detailed in Fadel Ali's report. This alignment between the experimental results and the simulation validates the design choices and reinforces the effectiveness of the soft actuator in achieving the desired performance. For further insights into the expansion metrics, please refer to Fadel Ali's report.



Figure 18 Simulation for liner part.

It was initially compared to the standard length of 127 mm.

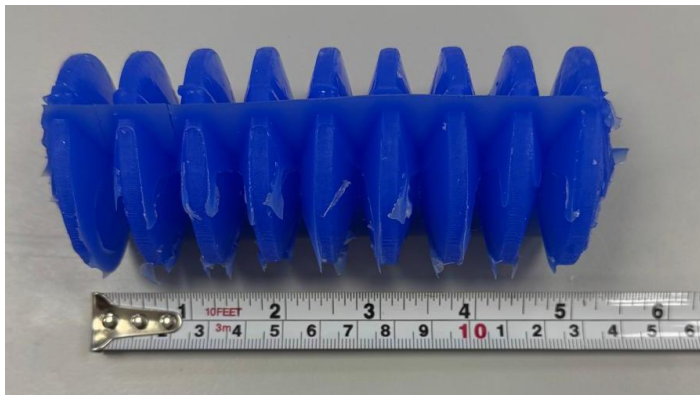


Figure 19 Initial Length of liner part.

After the apply the air , it expanded to approximately 203.2 mm, which is about 60% longer than its normal length.



Figure 20 Final Length of liner part.

$$\text{Expansion} = \frac{\text{Final Length} - \text{Initial Length}}{\text{Initial Length}} \times 100\%$$

$$\text{Expansion} = \frac{203.2 - 127}{127} * 100 = 60\%$$

Risks of the Project

1. Material Failure

The materials used may not withstand the stresses during casting or stretching, leading to structural failures.

2. Design Flaws

Inaccuracies in the design could result in unexpected outcomes, affecting the final piece's functionality or aesthetics.

3. Environmental Factors

Changes in temperature or humidity may impact the casting process and the materials' properties.

4. Time Delays

Unforeseen complications may cause delays in the project timeline, affecting deadlines and budgets.

5. Safety Hazards

The casting process involves high temperatures and potentially hazardous materials, posing risks to personnel.

6. Resource Availability

Limited availability of essential materials or tools may hinder progress and increase costs.

7. Technical Issues

Equipment malfunctions or technical difficulties could disrupt the production process.

8. Budget Overruns

Unexpected costs may arise, leading to budget overruns and financial strain on the project.

9. Regulatory Compliance

Failure to meet safety and quality regulations could result in legal issues or project delays.

10. Stakeholder Changes

Changes in stakeholder expectations or requirements may necessitate redesigns or additional work.

The project demonstrates a proactive approach to minimizing environmental impact through its design and intended manufacturing methods. The primary commitment lies in the exclusive use of PLA for the device's mechanical components, fabricated via 3D printing.

PLA is advantageous due to its biodegradability and generally lower environmental footprint compared to other common plastics like ABS, particularly concerning energy consumption during production and reduced off-gassing. The dedicated use of a 3D printer solely for PLA helps prevent cross-contamination with other plastic types, ensuring the material's integrity and facilitating its potential for future recycling efforts.

The eventual transition from continuous reliance on a power bank to direct wall power with an emergency battery backup is crucial for long-term energy efficiency. This change reduces the frequency of battery charging cycles and minimizes the overall energy footprint of the device. Furthermore, the strong potential for local manufacturing within Bahrain significantly reduces transportation-related carbon emissions typically associated with importing finished devices from international suppliers, contributing to a more localized and sustainable supply chain.

Future iterations of the device should critically explore the full lifecycle assessment (LCA). This comprehensive analysis would encompass environmental impacts from the initial stages of material sourcing and extraction, through energy consumption during manufacturing, transportation of components, and finally, end-of-life disposal and recyclability of the entire product. Such an assessment is vital for identifying areas where environmental performance can be further optimized.

The project also directly and indirectly contributes to several of the United Nations' Sustainable Development Goals (SDGs), while highlighting areas for further alignment.

Relevant Sustainable Development Goals (SDGs) for the Project

1. Responsible Consumption and Production (SDG 12)

Focuses on ensuring sustainable practices in the use of materials, such as the exclusive use of PLA, which is biodegradable and environmentally friendly.

2. Affordable and Clean Energy (SDG 7)

Emphasizes the transition to direct wall power and emergency battery backup, promoting energy efficiency and reducing reliance on non-renewable energy sources.

3. Industry, Innovation, and Infrastructure (SDG 9)

Highlights the use of 3D printing technology for local manufacturing, fostering innovation and sustainable industrial practices.

Discussion :

The development of the soft crawler robot for internal pipeline inspection has successfully realized its primary goals and objectives, demonstrating significant advancements in the field of soft robotics.

1. **Modular Design:** The objective of creating a modular robotic system was achieved, providing flexibility to adapt to various pipeline conditions. This adaptability is crucial for efficient inspections in confined spaces, allowing the robot to navigate through tight bends and curves effectively.
2. **Soft Actuator Integration:** The integration of soft actuators made from silicone materials has proven effective. The actuators demonstrated enhanced compliance and expansion capabilities, achieving expansion rates that closely matched simulation predictions. This success highlights the viability of soft robotics in addressing the limitations of traditional rigid inspection systems.
3. **Pneumatic Actuation System:** The development of a reliable pneumatic actuation system met the goal of enabling precise control over the robot's movements. The system effectively facilitated the inflation and deflation of actuators, crucial for achieving the desired locomotion within pipelines.
4. **Electronic Control Implementation:** The use of an Arduino-based electronic control system fulfilled the objective of providing real-time monitoring and operation. This control mechanism proved efficient, allowing for straightforward command inputs and reliable management of the robotic functions.
5. **Performance Evaluation:** The comprehensive testing and evaluation of the robot confirmed its functionality and performance in pipeline inspection tasks. Positive outcomes in terms of mobility, actuation reliability, and visual feedback underscore the robot's potential for practical applications.
6. **Cost-Effectiveness and Sustainability:** The project successfully addressed the goal of cost-effectiveness while prioritizing sustainability. Utilizing biodegradable materials like PLA and focusing on local manufacturing practices reflect a commitment to minimizing environmental impact.

Chapter 5: Conclusion

The development of the soft crawler robot for internal pipeline inspection marks a significant advancement in the use of soft robotics for complex inspection tasks. This project successfully achieved its core objectives, including the creation of a modular, adaptable system that enhances efficiency in navigating confined spaces.

The integration of silicone-based soft actuators demonstrated effective compliance and expansion, validating the design choices made throughout the development process. The reliable pneumatic actuation system provided precise control over movement, while the Arduino-based electronic control facilitated real-time operation and monitoring.

Testing results confirmed the robot's capability to perform effectively in various pipeline conditions, showcasing its potential for practical applications in the field. Additionally, the emphasis on cost-effectiveness and sustainability, through the use of biodegradable materials and local manufacturing practices, underscores the project's commitment to minimizing environmental impact.

Overall, this project not only contributes to the field of soft robotics but also sets the stage for future innovations and improvements. The insights gained and the successes achieved provide a strong foundation for further exploration and development, paving the way for more advanced and efficient pipeline inspection solutions.

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